Bram Grooten

Postdoc in Deep Learning



Postdoc at TU Eindhoven, researching the integration of **vision-language-action** modalities, **robotics**, reinforcement learning, and generalization. My long-term aim is to create reliable, adaptive, and intelligent agents acting in real-world scenarios. I pursue this by training VLAs for robotics, improving generalization under various environmental settings. Results so far include work at **Sony AI** (one policy for 100+ OOD vehicles) and an oral paper on the **UR5 robotic arm**.

bramgrooten.nl github scholar b.grooten@tue.nl

Work Experience

2026 – today **AMOR/e lab**, *Postdoctoral Researcher*, Netherlands
Working on multimodal foundation models for robotics within AMOR/e. Research member of the Eindhoven AI Systems Institute (EAISI).

2022 – 2026 **TU Eindhoven**, *PhD Candidate*, Netherlands
- Led projects on reinforcement learning and generalization (NeurIPS, ICLR, AAMAS).
- Co-applicant, MeluXina compute grant: 80,000 GPU hours (≈€104,000).
- Supervised/mentored 10+ MSc and BSc students; organised reading groups.

2024 – 2025 **Sony AI**, *Research Intern*, Zürich, Switzerland
Part of the Gran Turismo 7 (PlayStation) team for 7 months. The research led to an **oral** paper on RL generalization: a policy able to reliably race with 100+ OOD vehicles.

2023 **University of Alberta**, *Research Visit*, Edmonton, Canada
Visited the Intelligent Robot Learning lab for 5 months, joining Matthew Taylor at Amii. Our work on the UR5 robotic arm contributed to an accepted paper at AAMAS.

2017 **ThuisBaas**, *Research Intern*, Amsterdam, Netherlands
Analyzing the sound level of heat pumps and improving their solar energy model.

Education

2022 – 2026 **PhD Candidate**, *Eindhoven University of Technology*, Netherlands
Research focus: generalization and efficiency in deep RL/robotics via dynamic sparsity.

2018 – 2021 **Master Applied Mathematics**, *Eindhoven University of Technology (TU/e)*
Graduated cum laude. Thesis on multi-agent deep reinforcement learning for Hanabi.

2018 – 2021 **Master Science Education**, *Eindhoven University of Technology*
Acquired the official license to teach mathematics in Dutch high schools.

2014 – 2017 **Bachelor Applied Mathematics**, *Wentworth Institute of Technology & TU/e*
Studied abroad in Boston US, after which I continued in the Netherlands.

Teaching and Mentoring

2026 **Honors Academy Coach**, *AI track*, TU Eindhoven
Mentoring a group of excellent bachelor students on RL with the LeRobot SO-ARM101.

2026 **Supervisor**, *Giorgio Maioli*, MSc student
Honors Academy research project on World Action Models with language output

2026 **Supervisor**, *Madoc Karthaus*, MSc student
Semantic SLAM on a real Unitree Humanoid during his internship at ML6

2026 **Supervisor**, *Casper Adolfse*, MSc student
Project on Affective Robotics in collaboration with the University of Cambridge

2025 **Honors Academy Coach**, *AI track*, TU Eindhoven
Mentoring a group of excellent bachelor students on RL with the LeRobot SO-ARM101.

2025 **Supervisor**, *Quint Bakens*, MSc student
RL for Robotic bell-pepper picking (cum laude)

2024 **Supervisor**, *Camiel Oerlemans*, MSc student, NeurIPS workshop paper
Motion Prediction for Autonomous Driving (cum laude)

2023 **Supervisor**, *Mauk Muller*, MSc student
Skip Connections in Sparse Networks

2019 **Mathematics Teacher**, *Van Maerlantlyceum*, Eindhoven
At *Van Maerlant* I improved my teaching skills in classes of higher levels.

2019 **Mathematics Teacher**, *Maaslandcollege*, Oss
During the Education master I learned the teaching craft with this internship.

2018 **Guest Lecturer**, *ASML*, Netherlands
Bringing tech-enthusiasm to children with our guest lecture: *Make your own app!*

2017 **Exam Trainer**, *Lyceo*, Netherlands
Mathematics tutor for final year high school students.

2015 **Tutor**, *PBH Association at Harvard University*, Cambridge, MA, United States
Volunteering as a math tutor for children from the rough neighborhood of Mission Hill.

Selected Publications

2026 **B. Grooten**, J. Vanschoren. *How (and How Not) to Use Data Augmentation in VLA Post-Training*. NeurIPS'26 workshop RoboPAD: Post-Training Adaptation of Robot Foundation Models. link

2026 **B. Grooten**, P. MacAlpine, K. Subramanian, P. Stone, P.R. Wurman. *Out-of-Distribution Generalization with a SPARC: Racing 100 Unseen Vehicles with a Single Policy*. **Oral** at AAAI'26. link

2025 T. Tomilin, L. van den Boogaard, S. Garcin, **B. Grooten**, M. Fang, Y. Du, M. Pechenizkiy. MEAL: A Benchmark for Continual Multi-Agent Reinforcement Learning. Preprint. link

2025 **B. Grooten**, F. Hasanov, C. Zhang, Q. Xiao, B. Wu, Z. Atashgahi, G. Sokar, S. Liu, L. Yin, E. Mocanu, M. Pechenizkiy, D. Mocanu. *NeuroTrails: Training with Dynamic Sparse Heads as the Key to Effective Ensembling*. Preprint. link

2025 C. Muslimani, **B. Grooten**, D. Mamillapalli, M. Pechenizkiy, D. Mocanu, M. Taylor. *Boosting Robustness in Preference-Based Reinforcement Learning with Dynamic Sparsity*. AAMAS'25. link

2024 C. Oerlemans, **B. Grooten**, M. Braat, A. Alassi, E. Silvas, D. Mocanu. *LiMTR: Time Series Motion Prediction for Diverse Road Users through Multimodal Feature Integration*. NeurIPS'24 workshop Time Series in the Age of Large Models. link

2024 **B. Grooten**, T. Tomilin, G. Vasan, M. Taylor, A. Mahmood, M. Fang, M. Pechenizkiy, D. Mocanu. *MaDi: Learning to Mask Distractions for Generalization in Visual Deep Reinforcement Learning*. **Oral** at AAMAS'24. link

2024 **B. Grooten**. *Large Learning Agents: Towards Continually Aligned Robots with Scale in RL*. AAMAS'24 Doctoral Consortium. link

2023 A. Nowak, **B. Grooten**, D. Mocanu, J. Tabor. *Fantastic Weights and How to Find Them: Where to Prune in Dynamic Sparse Training*. NeurIPS'23. link

2023 **B. Grooten**, G. Sokar, S. Dohare, E. Mocanu, M. Taylor, M. Pechenizkiy, D. Mocanu. *Automatic Noise Filtering with Dynamic Sparse Training in Deep Reinforcement Learning*. Full-paper at AAMAS'23 & **Spotlight** at ICLR workshop SNN'23. link

2022 **B. Grooten**, J. Wemmenhove, M. Poot, J. Portegies. *Is Vanilla Policy Gradient Overlooked? Analyzing Deep Reinforcement Learning for Hanabi*. Adaptive and Learning Agents workshop at AAMAS'22. link

2021 **B. Grooten**. *Deep Reinforcement Learning for the cooperative card game Hanabi*. Master Thesis. link

2020 **B. Grooten**, B. Tulkens. *Programming in mathematics and physics classes*. Master Thesis. link

2020 **B. Grooten**, I. Schilstra, W. van der Hert, D. van Genuchten. *AI Snakes Competition*. Technical Report. link

2017 **B. Grooten**. *Weighted bipartite b-matching for Amsterdam high schools in practice*. Bachelor Thesis. link

Invited Talks

2026 **Leibniz University Hannover**, *Adaptive RL: Generalization in Robotics*
Workshop on RL for the Real World. Hannover, Germany. Aug 10th.

2026 **Openmind Winter School**, *Lean & Adaptive Agents for Generalization*
Universiti Tunku Abdul Rahman. Selangor, Malaysia. Jan 27th.

2025 **EAISI**, *How to Make Sure your Robot is Not Distracted*
Eindhoven AI Systems Institute, Netherlands. Dec 17th.

2025 **VU Amsterdam**, *Lean & Adaptive Agents for Generalization in Deep RL*
Robotics lab. Amsterdam, Netherlands. Dec 9th.

2025 **TU Darmstadt**, *Lean & Adaptive Agents for Generalization in Deep RL*
Jan Peters' IAS lab. Darmstadt, Germany. Dec 5th.

2025 **TU Delft**, *Lean & Adaptive Agents for Generalization in Deep RL*
Delft, Netherlands. Dec 4th.

2024 **Massachusetts Institute of Technology**, *Efficient Focus for Autonomous Agents*
LIDS: Cathy Wu's lab. Cambridge, MA, United States. Aug 19th.

2024 **Sony AI**, *Efficient Focus for Autonomous Agents: Generalization in Deep RL*
Tech Talk Series. Zürich, Switzerland. Jul 24th.

2024 **ETH Zürich**, *Efficient Focus for Autonomous Agents: Generalization in Deep RL*
Computational Robotics Lab. Zürich, Switzerland. Jul 17th. Announcement.

2024 **ML Collective**, *Efficient Focus for Autonomous Agents: Generalization in Deep RL*
Online reading group by Rosanne Liu: "DL: Classics and Trends." Mar 15th. Website.

2024 **Leiden University**, *Efficient Focus for Autonomous Agents*
Leiden, Netherlands. Feb 13th.

2023 **University of Calgary**, *Efficient Focus for Autonomous Agents*
Calgary AB, Canada. Oct 25th. Website.

2023 **LIFE at MIT**, *MaDi: Learning to Mask Distractions from Pixels*
Online reading group "Learning in Foundation Environments." Oct 23rd.

2023 **University of Alberta**, *Efficient Focus for Autonomous Agents*
Edmonton AB, Canada. Aug 25th. Website. Recording.

2023 **PyData**, *Automatic Noise Filtering*
Eindhoven, Netherlands. Apr 26th. Announcement.

2023 **ProDrive**, *Automatic Noise Filtering*
Eindhoven, Netherlands. Feb 16th.

2022 **Jagiellonian University**, *Efficient AI for Autonomous Agents*
Kraków, Poland. Jul 5th.

Honors and Awards

2026 Oral paper at AAAI

2025 Finalist in the *Talking Science* pitch competition.

2024 Oral paper at AAMAS

2024 Scholarship recipient at AAMAS

2023 Spotlight paper: Sparse Neural Networks workshop at ICLR

2023 Scholarship recipient at AAMAS

2021 Cum laude MSc graduation, with a 9/10 for my thesis on *Deep Reinforcement Learning for the cooperative card game Hanabi*

2020 Winner of the international Angry Birds Level Generation contest

2020 Silver medal in the international AI Snakes competition

2018 ASML Technology Scholarship: Awarded to 25 master's students annually

2015 Dean's list: Top student at Wentworth Institute of Technology

2014 Bèta award: Student with highest grades in all STEM courses

2014 Cum laude high school graduation

In the News

2026 "Eindhovens algoritme leert robots generaliseren" High-Tech Systems.

2026 "Dat moet te automatiseren zijn" Eindhovens Dagblad.

2026 "Home stretch" Cursor article.

2026 "A handy robot that doesn't get distracted" TU Eindhoven article.

2025 "Winner Talking Science 2025" Cursor article.

Grants and Funding

2026 **Large Compute Grant**, *Snellius*, The Netherlands
Co-applicant on a successful grant for 45,580,000 SBU credits on Snellius.
Amount: 680,000 GPU hours (≈ €850,000)

2025 **NWO SURF Grant**, *Snellius*, The Netherlands
Personal application EINF-12945 for 1 million SBU credits on the Netherlands' largest HPC.
Amount: 15,000 GPU hours (≈ €19,500)

2024 **LuxProvide Compute Grant**, *MeluXina HPC*, Luxembourg
Co-applicant on a successful grant to Luxembourg's largest compute cluster.
Amount: 80,000 GPU hours (≈ €104,000)

2023 **Personal Research Grant**, *University of Alberta*, Canada
Allowance for a 5-month visiting scholar position at the Intelligence Robot Learning lab.
Amount: $7000

2022 **NWO SURF Grant**, *Snellius*, The Netherlands
Personal application EINF-3098 for 1 million SBU credits on the Netherlands' largest HPC.
Amount: 15,000 GPU hours (≈ €19,500)

Projects

2023 **Mila**, *Montreal*, Canada, dlrl.ca
Accepted at the DLRL summer school held at Mila.

2022 **European Summer Schools**, *Kraków, Vilnius, Milan*
Participated in three machine learning summer schools: MLSS, EEML, and M2L.

2021 **Serpentine AI**, serpentine.ai
Chairman of the student team which develops AI for e-Sports. Led the team through many international AI programming competitions while maintaining shared codebases.

2020 **Angry Birds Competition**, aibirds.org
Winning team in this challenging level generation contest.

2020 **AI Snakes Competition**, Technical Report
Team-lead of a prize-winning group of developers.

2019 **MIT Battlecode**, battlecode.org
Programming competition hosted by MIT, where we reached the top 30.

Community

Reviewing: ECML-PKDD 2026, CATS ICML Workshop 2026, ICML 2025, EWRL 2025, AAAI 2025, NeurIPS 2024, Nature Communications 2024, SNN Workshop 2023, 2022

Organizing: Diffusion lecture series 2026, DAI Seminar series 2025, 2024, IJCAI Tutorial on Sparse Training 2023

Volunteering: HPC Administrator 2024, PhD Council member 2023, Math tutor at the Phillips Brooks House Association of Harvard University 2015